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Data Communication CS601 Lecture No 14

Lecture Prepared By

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Today's Topic

Analog to Digital Conversion

1. (PCM) - Pulse Code Modulation
    - i) Sampling
    - ii) Quantization
    - iii) Encoding
2. (DM) - Delta Modulation

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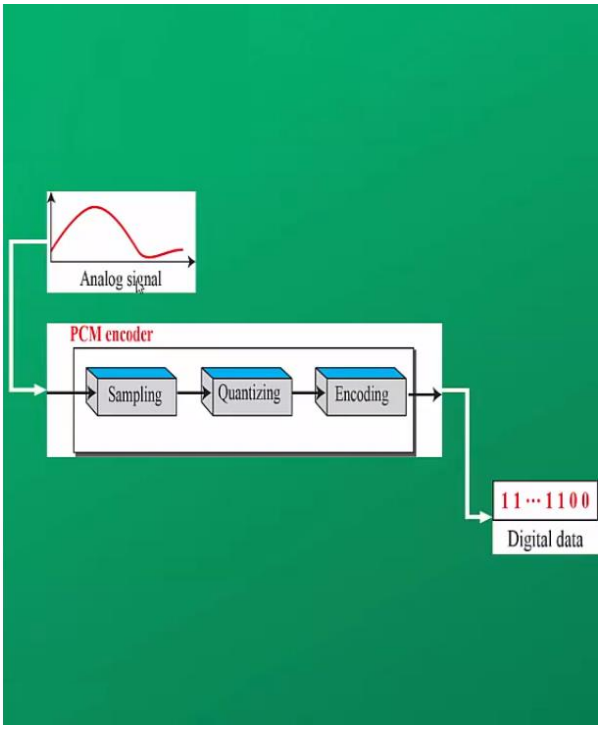


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Analog to Digital Conversion

In this method we convert the analog signals to the digital signals. In this method first of all we convert the data into Digitization than convert it into Digital Signals. Its mean that two process are involved in it.

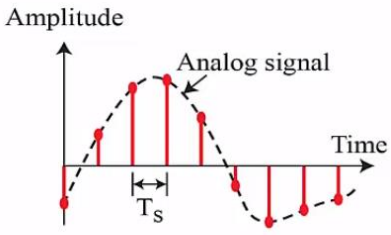
- PCM consists of three steps to digitize an analog signal:
  1. (PCM) - Pulse Code Modulation
     -- Sampling (Pulse Amplitude Modulation)
     -- Quantization
     -- Encoding
  2. (DM) - Delta Modulation



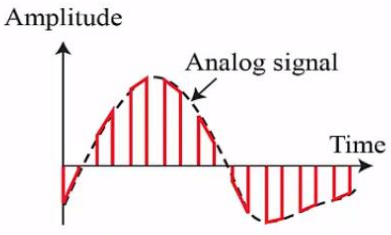
Sampling

Series of pulses with amplitude values between min and max signal amplitude.

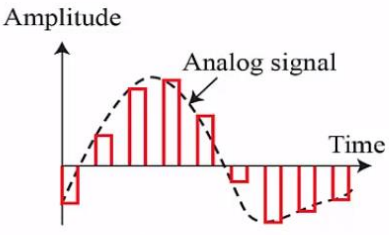
Three Different Sampling Methods for PCM



a. Ideal sampling



b. Natural sampling



c. Flat-top sampling

TS → Sampled Interval or Sample Period

It has different pluses at different Amplitude. It is very difficult to deploy in Hardware.

It has high speed Switch. It has very small time interval

It is also called sample and hold technique.

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Sample is a piece of data taken from the whole data which is continuous in the time domain. When a source generates an analog signal and if that has to be digitized, having 1s and 0s i.e., High or Low, the signal has to be discretized in time. This discretization of analog signal is called as Sampling.

نمونہ پورے ڈیٹا سے لیا گیا ڈیٹا کا ایک ٹکڑا ہے جو ٹائم ڈومین میں مسلسل ہوتا ہے۔ جب کوئی ذریعہ ایک یعنی ہائی یا لو ہو، تو سگنل کو 1s اور 0s اینالاگ سگنل تیار کرتا ہے اور اگر اسے ڈیجیٹائز کرنا ہو، جس میں وقت کے ساتھ الگ کرنا ہوگا۔ ینالاگ سگنل کی اس ڈسکریٹائزیشن کو سیمپلنگ کہا جاتا ہے۔

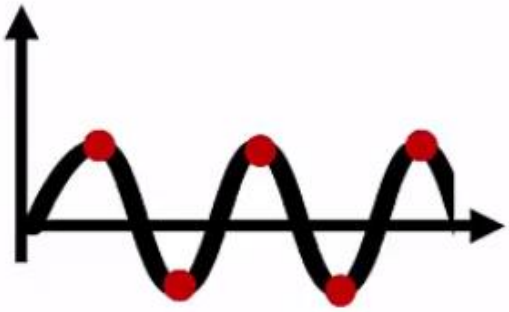
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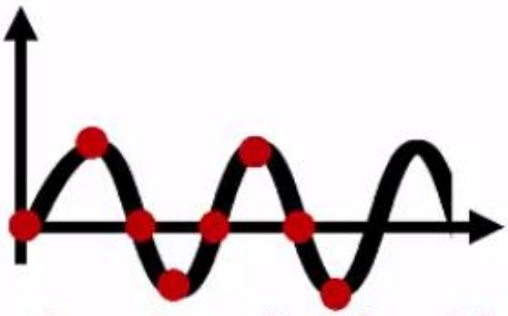
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Nyquist Sampling Rate

Nyquist → fs (Frequency of Sampling) = 2fh (Twice Heights Frequency)



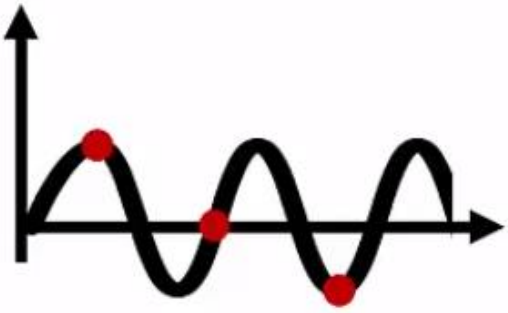
Ideal Sampling $f_s = 2\,f$



Over Sampling $f_s = 4\,f$

(Double Nyquist Rate)

Extra on Line Sample has no Value



Under Sampling $f_s = f$

(Half Nyquist Rate)

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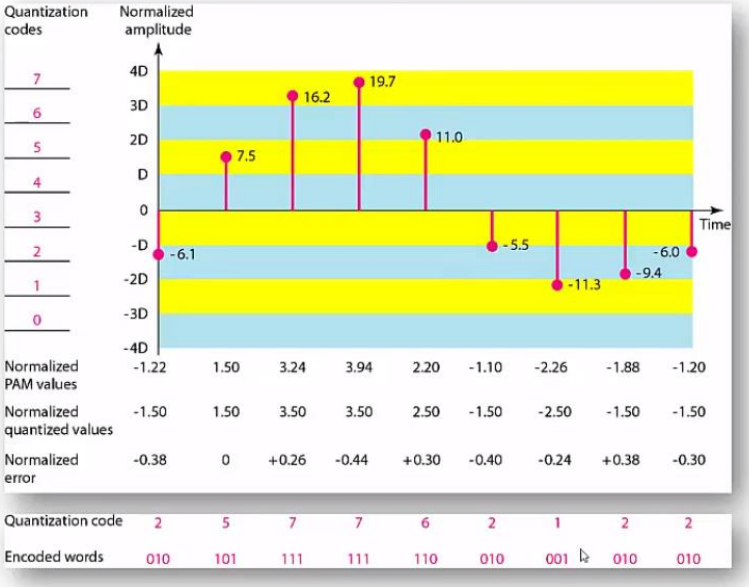
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Quantization

Quantization is normalization of sample data according to certain level. The no. of level chose on two factors.

-- Range of Amplitude

-- Accuracy (If we acquire higher accuracy than we need more Quantization Level and Voice Versa).



Encoding

The digitization of the analog signal is done by the encoder. After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an n bit code. Encoding also minimizes the bandwidth used.

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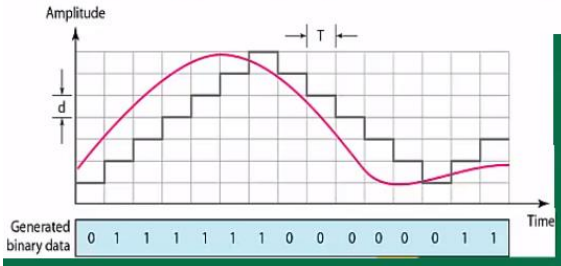


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Delta Modulation

This scheme sends only the difference between pulses, if the pulse at time t_{n+1} is higher in amplitude value than the pulse at time t_n,

- Then a single bit, say a "1", is used to indicate the positive value.
- If the pulse is lower in value, resulting in a negative value, a "0" is used.
- This scheme works well for small changes in signal values between samples.
- If changes in amplitude are large, this will result in large errors.



Delta Modulation

- Delta modulation is a simpler technique
- DM finds the change from the previous sample
- No code words